Assignment on Variable Selection

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In [12]:
```
# read data in
df <-read.csv("C:/Users/tyler/OneDrive/Documents/GitHub/work_to_show/Classes Spring
head(df)
colnames(df)
```

A data.frame: 6 × 9

	Ht	Wt	yd40	Vertical	Bench	BroadJump	Cone3	Sl
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<
1	-0.05226864	-0.8310177	-0.9919680	0.7947617	-0.1775535	0.5162616	-1.5108623	-1.29
2	-2.14835258	-1.2441244	-0.4241502	0.6746024	-1.2017744	-0.3326489	-0.6004284	-0.26
3	0.99577333	1.6911069	1.6263029	-0.8874689	1.1393018	-1.1815595	1.6366376	1.59
4	1.34791543	1.2780002	1.8471210	-1.1277876	1.2856191	-1.7121286	2.3129599	0.97
5	1.69586537	1.6911069	2.2887570	-1.0076283	-0.6165053	-1.1815595	-0.7044780	2.70
6	-1.79621048	-1.2223819	-0.8026954	0.5544431	-0.1775535	0.9407169	-0.4443541	-0.55

'Ht' · 'Wt' · 'yd40' · 'Vertical' · 'Bench' · 'BroadJump' · 'Cone3' · 'Shuttle' · 'Draftedyes'

In [9]:
```
# Stepwise Regression using AIC
# install.packages("MASS")
library(MASS)
step_model <- stepAIC(lm(Draftedyes ~ ., data = df), direction = "both", trace = FA
# Print the selected variables
print(step_model$anova)
```

```
Stepwise Model Path 
Analysis of Deviance Table

Initial Model:
Draftedyes ~ Ht + Wt + yd40 + Vertical + Bench + BroadJump + 
    Cone3 + Shuttle

Final Model:
Draftedyes ~ Wt + yd40 + Cone3


         Step Df   Deviance Resid. Df Resid. Dev       AIC
1                                 258   48.26835 -438.6962
2  - Vertical  1 0.06585749       259   48.33420 -440.3322
3   - Shuttle  1 0.10060691       260   48.43481 -441.7770
4        - Ht  1 0.11008586       261   48.54490 -443.1708
5     - Bench  1 0.05452453       262   48.59942 -444.8711
6 - BroadJump  1 0.32655477       263   48.92598 -445.0830
```

What to pick for Stepwise Regression using AIC

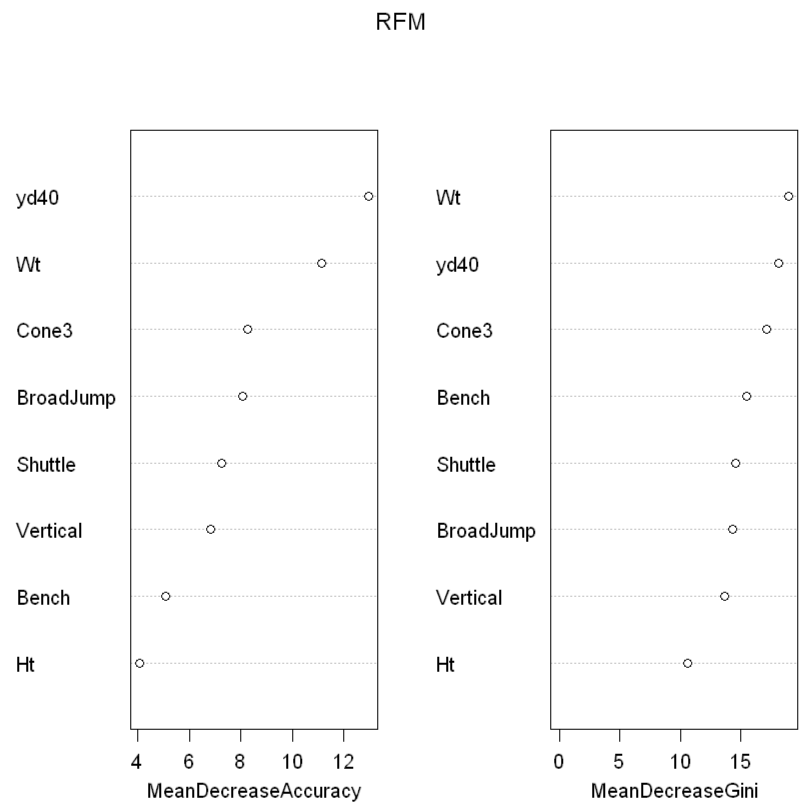
Predictors I would pick: Wt yd40 Cone3

For a regression model to predict Draftedyes I would use the predictors Wt, yd40, Cone3. My model would be "Draftedyes ~ Wt + yd40 + Cone3".

In [13]:
```
# Random Forest
library(randomForest)
df$Draftedyes<-factor(df$Draftedyes)
RFM=randomForest(Draftedyes ~.,data=df, importance=TRUE)
importance(RFM)
varImpPlot(RFM)
```

A matrix: 8 × 4 of type dbl

	0	1	MeanDecreaseAccuracy	MeanDecreaseGini
Ht	0.8577694	4.757633	4.079098	10.61858
Wt	7.1899436	6.341123	11.131492	19.05648
yd40	6.3441233	10.537335	12.980610	18.16746
Vertical	5.5116114	2.418433	6.838149	13.69633
Bench	-1.5117808	9.036706	5.089367	15.55658
BroadJump	10.2936505	-2.594324	8.071412	14.38485
Cone3	4.9763782	4.756804	8.251325	17.20850
Shuttle	6.9825320	1.455912	7.241367	14.59925



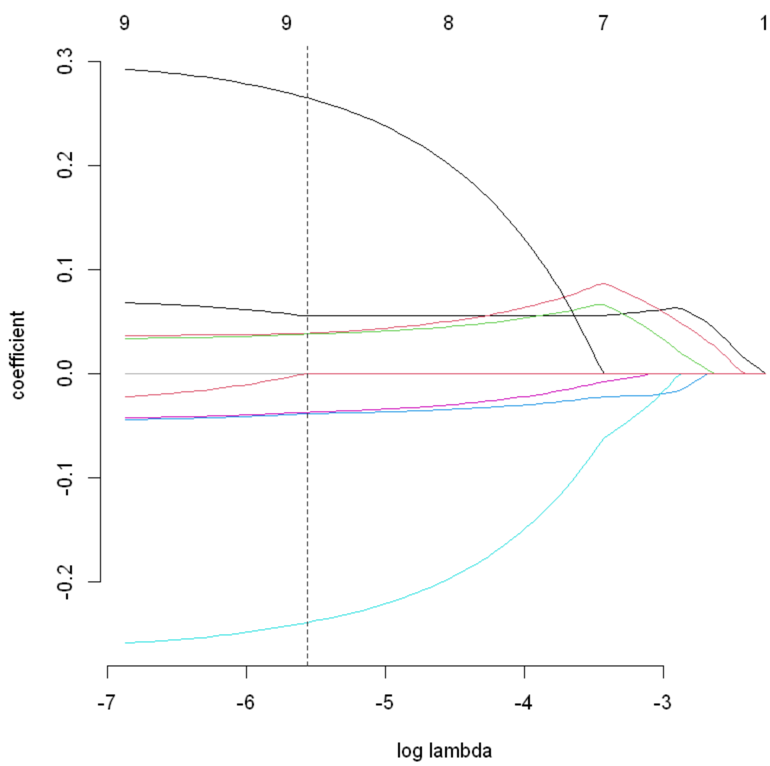
What to pick for Random Forest

Predictors I would pick: Wt, yd40, Broadjump, Cone3

For a regression model to predict Draftedyes I would use the predictors Wt, yd40, BroadJump, Cone3. My model would be "Draftedyes ~ Wt + yd40 + BroadJump + Cone3". I chose this because the %IncMSE < 5.

In [11]:
```
# Lasso Regression
# Note these variables are all on the same scale, so no standardization is needed.
# install.packages("gamlr")
library(gamlr)
modMat<-model.matrix(~Ht + Wt + yd40 + Vertical + Bench + BroadJump +
    Cone3 + Shuttle, data=df)
fit<-gamlr(x=modMat,y=df$Draftedyes)
plot(fit)
coef(fit)
```

```
10 x 1 sparse Matrix of class "dgCMatrix"
                   seg72
intercept    1.36329588
(Intercept)  .
Ht           0.03816811
Wt           0.26472550
yd40        -0.23852531
Vertical     .
Bench        0.03886583
BroadJump    0.05558262
Cone3       -0.03811878
Shuttle     -0.03680288
```



What to pick for Lasso Regression

Predictors I would pick: Ht, Wt, yd40, Bench, BroadJump, Cone3, Shuttle.

For a regression model to predict Draftedyes I would use the predictorsHt, Wt, yd40, Bench, BroadJump, Cone3, Shuttle. My model would be "Draftedyes ~ Ht+Wt+yd40+Bench+BroadJump+Cone3+Shuttle.". I chose this because the lasso regression chose these but would not take Vertical.

Models I would pick

I would choose stepwise model because of landing on 3 predictors. The lasso regression got better as all the other variables got removed. The random forest is the second best in my opinion, based off it being a percentage of it being removed and the increase of MSE%.

```
In [ ]:
```